

Google Cloud

Dashboard

Explore

Paths

Google Cloud Skills Boost for Partners

Course > Build Custom Processors with Document AI >

Build Custom Processors with Document AI

Course · 6 hours 83% 15 minutes complete

Course overview

Build Custom Processors with Document AI

Optical Character Recognition (OCR) with Document AI (Python)

Form Parsing with Document AI (Python)

Using Specialized Processors with Document AI (Python)

Uptraining with Document AI Workbench

Custom Document Extraction with Document AI Workbench

Quick tip: Review the prerequisites before you run the lab

End Lab

01:03:59

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-b4-b0c43387a53bi

Password

IDDwTC0eh0jV

Google Cloud Project ID

qwiklabs-gcp-03-0626d4b

Lab

1 hour 30 minutes

No cost

Intermediate

★★★★★

This lab may incorporate AI tools to support your learning.

GSP513

Google Cloud Self-Paced Labs

Lab instructions and tasks

80/100

GSP513

Introduction

Challenge scenario

Task 1. Enable the Document AI API

Task 2. Create a processor

Task 3. Define processor fields

Task 4. Import a document into a dataset

Task 5. Label a document

Task 6. Assign an annotated document to the training set

Task 7. Import pre-labeled data to the training and test sets

Task 8. Train the

Introduction

In a challenge lab you're given a scenario and a set of tasks. Instead of following step-by-step instructions, you will use the skills learned from the labs in the course to figure out how to complete the tasks on your own! An automated scoring system (shown on this page) will provide feedback on whether you have completed your tasks correctly.

When you take a challenge lab, you will not be taught new Google Cloud concepts. You are expected to extend your learned skills, like changing default values and reading and researching error messages to fix your own mistakes.

To score 100% you must successfully complete all tasks within the time period!

This lab is recommended for students who have enrolled in the [Build Custom Processors with Document AI: Challenge Lab](#) course. Are you ready for the challenge?

Before you click the Start Lab button

starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud account, you may incur charges to that account.

Challenge scenario

You were onboarded at Cymbal Labs just a few months ago. Cymbal Labs is a leading bioscience firm dedicated to advancing innovation in biotechnology. Cymbal Labs researches, manufactures, and distributes a wide variety of healthcare solutions across a range of medical disciplines, including internal medicine, oncology, immunology, and cardiology.



By harnessing Document AI Workbench, a powerful tool for creating custom document processors, Cymbal Labs accelerates its discoveries and gains valuable insights from

process, identify licensing and collaboration opportunities, and manage intellectual property relating to its own patents.

Your team has been working to create a Custom Document Extractor that can extract key information from public patent documents. Your job is to give them a hand and help them get their Document AI workflows up and running. As part of this demonstration, they have a list of tasks they would like to see you do in an allotted period of time in a sandbox environment.

Your challenge

Your tasks include the following:

- Create a custom processor.
- Import a document into a dataset.
- Define processor schema.
- Label a document.
- Assign an annotated document to the training set.
- Import pre-labeled data to the training and test sets.
- Kick off a training job.

Task 1. Enable the Document AI API

Before you can begin using Document AI, you must enable the API.

1. Enable the Document AI API.



Enable the Document AI API

Check my progress

Assessment Completed!

Task 2. Create a processor

You must first create an Custom extractor processor to use for this task.

1. Create a **Custom Extractor** processor with the name `custom-doc-extractor-0f9b`. Use the region **US (United States)**.



Create a Processor

Check my progress

Assessment Completed!

Note: The processor takes a few minutes to create and is ready once the dataset has been initialized.

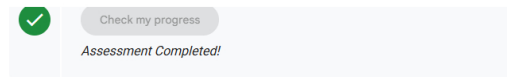
Task 3. Define processor fields

labels that you use to annotate documents.

1. Create each of the following labels for the processor schema.

Name	Data Type	Occurrence
applicant_line_1	Plain Text	Required once
application_number	Number	Required once
class_international	Plain Text	Required once
class_us	Plain Text	Required once
filing_date	Datetime	Required once
inventor_line_1	Plain text	Required once

issuer	Plain text	Required once
patent_number	Number	Required once
publication_date	Datetime	Required once
title_line_1	Plain text	Required once



Task 4. Import a document into a dataset

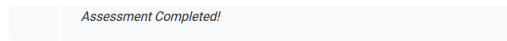
Next, you import a test document into your dataset.

1. Import the following document into your dataset:

```
gs://cloud-samples-  
data/documentai/codelabs/challenge/unlabeled/us_001.pdf
```



Import a Test Document



Task 5. Label a document

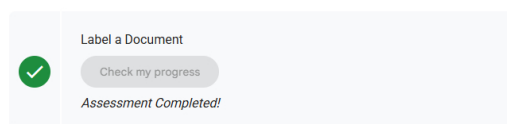
The process of selecting text in a document, and applying labels is known as annotation. In this section, you will annotate a document with the labels you defined in the previous section.

The generative AI model should have populated most of the labels for you. You can use and edit the suggested labels as needed.

1. Add the following labels to the `us_001` document:

- **applicant_line_1** = Colby Green
- **application number** = 679.694
- **filing_date** = Aug. 17, 2017
- **inventor_line_1** = Colby Green
- **issuer** = US
- **patent_number** = 10,136,408
- **publication_date** = Nov. 20, 2018
- **title_line_1** = DETERMINING HIGH VALUE

The labeled patent document should look like this when complete:

[illegible]

Task 6. Assign an annotated document to the training set

document to the training set

Now that you have labeled this example document, you can assign it to the training set.

1. Assign the `us_001` document to the training set.

Assign Document to Training Set

Assessment Completed!

Task 7. Import pre-labeled data to the training and test sets

1. Import the following pre-labeled documents into your dataset. For Data split, use **Auto-split**. Leave *Import with auto-labeling unchecked*.

`gs://cloud-samples-data/documentai/codelabs/challenge/labeled`

Import pre-labeled data

Check my progress

Task 8. Train the processor

Now that you have imported the training and test data, you can kick off a training job for the processor.

1. Train a new version of the processor. Use the following name for the processor version: `cde-version-1`.

Train the Processor

Check my progress

Assessment Completed!

Congratulations!

Congratulations! In this lab you verified your skills on Document AI Workbench by creating a custom processor and dataset, importing documents, labeling documents, and training a processor. You can now use Document AI Workbench to create custom processors for your own use cases.



Build Custom Processors with Document AI

Machine Learning & AI

SKILL BADGE · INTERMEDIATE

Earn your next skill badge

This self-paced lab is part of the [Build Custom Processors with Document AI](#) course. Completing this skill badge quest earns you the badge above, to recognize your achievement. Share your badge on your resume and social platforms, and announce your accomplishment using #GoogleCloudBadge.

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

Manual Last Updated April 18, 2024

Lab Last Tested April 18, 2024

Copyright 2025 Google LLC. All rights reserved. Google and the Google logo are trademarks of Google LLC. All other company and product names may be trademarks of the respective companies with which they are associated.

Build Custom 